

QA Tester Occupancy

Capacity & Occupancy calculation guide | App v1.0.4

Current assignments contribute lifecycle-weighted points. 8 points equal 100% occupancy. Completed requests remain visible for tracking and contribute zero occupancy points.

1. The formula

Occupancy % = $\text{round}((\text{total current assignment points} / 8) \times 100)$

- Only requests in a current tester or analyst workload status contribute points.
- Functional Execution in Progress contributes 2.50 points per request.
- Functional and Performance requests divide their points equally among assigned testers.
- SAST and DAST use one assigned Security Analyst; their weights are not divided.
- The system rounds the total points to two decimals, then the percentage to a whole number. Exact .5 percentage ties round to the nearest even whole number.

2. Capacity bands

Occupancy	Band	Meaning
0%	Available	No weighted current assignments.
1-49%	Light	Capacity is available.
50-79%	Balanced	Active workload below the high threshold.
80-99%	High	Approaching planned capacity.
100%	Full	At planned capacity: 8 points.
>100%	Overloaded	Assignments exceed planned capacity.

3. Assignment weights

Weights apply to the request before any shared-tester division. Only Functional Execution in Progress has changed to 2.50; other lifecycle weights retain their configured values.

Functional		Performance		SAST and DAST	
Status	Points	Status	Points	Status	Points
Tester Assigned	0.50	Environment Setup	1.00	Configuration	0.75
Test Design	1.00	Script Development	1.00	Scanning	1.00
Execution In Progress	2.50	Baseline	0.75	Finding Validation	0.75
Defect Raised	0.50	Load Test Execution	1.00	Remediation	0.50
Waiting For Fix	0.00	Result Analysis	0.25	Assigned To Requester	0.10
Retesting	0.75	Defect/Fix/Retest	0.75	Waiting For Fix	0.00
QA Completed	0.15	Report	0.15	Assigned To Lead	0.10
QA Change Review	0.15	Sign-off Pending	0.10	Rescan	0.75
QA Sign-off Pending	0.10			Security Complete	0.15
QA Signed Off	0.10			Report Ready	0.10
Requester Verification	0.05				

4. Worked examples

Example 1 - One Functional execution request

Assignment	Calculation	Points
Functional: Execution in Progress	2.50	2.50

Total: 2.50 points. Occupancy = $(2.50 / 8) \times 100 = 31\%$.

The exact occupancy is 31.25%; the dashboard displays 31% (Light).

Example 2 - Mixed Functional and Performance work

Assignment	Calculation	Points
Functional: Execution in Progress	2.50	2.50
Functional: Retesting	0.75	0.75
Performance: Baseline	0.75	0.75
Performance: Result Analysis	0.25	0.25

Total: 4.25 points. Occupancy = $(4.25 / 8) \times 100 = 53\%$.

The new total is 4.25 points and 53% occupancy (Balanced).

Example 3 - Functional execution shared by two testers

Assignment	Calculation	Points
Functional: Execution in Progress	$2.50 / 2$ testers	1.25

Total: 1.25 points. Occupancy = $(1.25 / 8) \times 100 = 16\%$.

Each tester receives 1.25 points and 16% occupancy. Shared Performance requests follow the same division rule.

Example 4 - Waiting work consumes zero points

Assignment	Calculation	Points
Functional: Waiting for Fix	0.00	0.00
SAST: Waiting for Fix	0.00	0.00
DAST: Scanning	1.00	1.00

Total: 1.00 points. Occupancy = $(1.00 / 8) \times 100 = 12\%$.

Waiting requests remain in workload counts. Security scanning still contributes 1.00 point; 12.5% rounds to 12%.

Example 5 - Near-complete work

Assignment	Calculation	Points
Functional: QA Completed	0.15	0.15
Performance: Sign-off Pending	0.10	0.10
SAST: Report Ready	0.10	0.10

Total: 0.35 points. Occupancy = $(0.35 / 8) \times 100 = 4\%$.

Near-complete work keeps a small weight because follow-up may still be required.

Example 6 - Full capacity

Assignment	Calculation	Points
Three Functional execution requests	3×2.50	7.50
Functional: Tester Assigned	0.50	0.50

Total: 8.00 points. Occupancy = $(8.00 / 8) \times 100 = 100\%$.

7.50 execution points plus 0.50 queued points equal 8.00 points: 100% (Full).

Example 7 - Overloaded tester

Assignment	Calculation	Points
Four Functional execution requests	4×2.50	10.00

Total: 10.00 points. Occupancy = $(10.00 / 8) \times 100 = 125\%$.

Occupancy can exceed 100%. The dashboard shows 125% (Overloaded); available capacity is 0%.

5. How the date filter behaves

- The period filters completed-work history: All time, Last 3 days, Last 15 days, Last month, or custom From/To dates.
- Current assignments remain visible even if they were created before the selected period.
- Changing the date filter changes completed-request history, not current occupancy.

6. Reading the dashboard correctly

Dashboard value	Meaning
Assigned	Number of current requests attributed to the tester or analyst.
Active Work	Current assignments not classified as queued, waiting, or near complete.
Queued	Functional requests at Tester Assigned.
Waiting / Blocked	Requests waiting on fixes or assigned back to a requester.
Near Complete	Completion, report, sign-off, or requester-verification stages.
Available Capacity	100 minus occupancy, never below 0%.
Request tracking	Current assignments plus completed requests in the selected period.

7. Interpretation notes

- Occupancy is a planning indicator, not a timesheet or a measure of individual performance.
- Weights estimate concurrent workflow demand; they do not represent hours worked.
- Many low-weight near-complete items do not equal the same number of execution assignments.
- QA Leads should review the underlying request ledger before assigning work.

Calculation source: backend/app/routers/dashboard.py - TESTER_CAPACITY_POINTS and the Functional, Performance, and Security load maps. Generated from the system configuration on 2026-10-08.